

LAB EXERCISE

Priority Queue & Binary Heaps



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QUESTIONS 10 / 10 completed

Max Heap

MAX HEAP

"Write a C program to perform the following operations:
Insert the employee salaries: 45000, 52000, 61000, 48000, 75000, 68000, 59000. Display the Max Heap after every insertion
Delete the highest salary from the heap. Display the heap after deletion"

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 #define MAX 100
4
5 int heap[MAX];
6 int size = 0;
7
8 void insert(int value) {
9     int i = size;
10    heap[size++] = value;
11
12    while (i != 0 && heap[(i - 1) / 2] < heap[i]) {
13        int temp = heap[i];
14        heap[i] = heap[(i - 1) / 2];
15        heap[(i - 1) / 2] = temp;
16        i = (i - 1) / 2;
17    }
18 }
19
20 void display() {
21     int i;
```

INPUT

Enter number of employees: 7

Enter salaries:
45000
52000
61000
48000
75000
68000
59000



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QUESTIONS 10 / 10 completed

Final Priority Queue

FINAL PRIORITY QUEUE

"Write a C program to implement the following operations:
Insert student requests with priorities: (101, 3), (102, 1), (103, 5), (104, 2), (105, 4)
Display the queue elements in priority order
Delete the highest priority request"

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int id[100], priority[100];
4 int n;
5
6 void sortQueue()
7 {
8     int i, j, temp;
9
10    for(i = 0; i < n - 1; i++)
11    {
12        for(j = i + 1; j < n; j++)
13        {
14            if(priority[i] < priority[j])
15            {
16                temp = priority[i];
17                priority[i] = priority[j];
18                priority[j] = temp;
19
20                temp = id[i];
21                id[i] = id[j];
```

OUTPUT

Enter number of requests: Enter Request ID and Priority:

INPUT

4
101 3
102 1
103 5
104 2

← → ↻

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QUESTIONS 10 / 10 completed

Heap Sort

HEAP SORT

"Write a C program to implement Heap Sort for the following set of numbers:
45, 12, 78, 34, 23, 89, 67, 10, 56, 99
The program should:
Construct the Max Heap
Display the heap after each adjustment
Sort the elements in ascending order using Heap Sort"

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 void heapify(int arr[], int n, int i)
4 {
5     int largest = i;
6     int left = 2 * i + 1;
7     int right = 2 * i + 2;
8     int temp;
9
10    if(left < n && arr[left] > arr[largest])
11        largest = left;
12
13    if(right < n && arr[right] > arr[largest])
14        largest = right;
15
16    if(largest != i)
17    {
18        temp = arr[i];
19        arr[i] = arr[largest];
20        arr[largest] = temp;
21    }
```

OUTPUT

INPUT

7
45 12 78

← → ↻

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Heap and Priority Queue

HEAP AND PRIORITY QUEUE

Write a C program to implement all Heap and Priority Queue operations through a menu-driven program."

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int arr[100];
4 int n;
5
6 void sortDescending()
7 {
8     int i, j, temp;
9
10    for(i = 0; i < n - 1; i++)
11    {
12        for(j = i + 1; j < n; j++)
13        {
14            if(arr[i] < arr[j])
15            {
16                temp = arr[i];
17                arr[i] = arr[j];
18                arr[j] = temp;
19            }
20        }
21    }
```

OUTPUT

INPUT

7
45 12 78 34 23 90 11
3

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QUESTIONS 10 / 10 completed

Heap Property

HEAP PROPERTY

Write a C program to check whether a given binary tree satisfies Heap property.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int arr[100], n;
4
5 int isMaxHeap()
6 {
7     int i;
8
9     for(i = 0; i <= (n - 2) / 2; i++)
10    {
11        if(2 * i + 1 < n && arr[i] < arr[2 * i + 1])
12            return 0;
13
14        if(2 * i + 2 < n && arr[i] < arr[2 * i + 2])
15            return 0;
16    }
17
18    return 1;
19 }
20
21 int main()
```

OUTPUT

INPUT

7
90 70 50 40

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QUESTIONS 10 / 10 completed

Heap Operations

HEAP OPERATIONS

"Write a menu-driven C program to implement all Heap operations for the following elements: 18, 25, 40, 12, 30, 50, 10, 45 and perform the following operations: Heap creation, Insertion, Deletion, Heap display and Searching an element"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int heap[100];
4 int size = 0;
5
6 void insert(int value)
7 {
8     int i = size;
9     heap[size] = value;
10    size++;
11
12    while(i > 0)
13    {
14        int parent = (i - 1) / 2;
15
16        if(heap[parent] < heap[i])
17        {
18            int temp = heap[parent];
19            heap[parent] = heap[i];
20            heap[i] = temp;
21            i = parent;
22        }
23    }
24 }
```

OUTPUT

Heap after creation:

INPUT

55

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Merging Heaps

MERGING HEAPS

Write a C program to merge two Max Heaps into a single Heap.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int heap[100];
4 int n;
5
6 void heapify(int i)
7 {
8     int largest = i;
9     int left = 2 * i + 1;
10    int right = 2 * i + 2;
11    int temp;
12
13    if(left < n && heap[left] > heap[largest])
14        largest = left;
15
16    if(right < n && heap[right] > heap[largest])
17        largest = right;
18
19    if(largest != i)
20    {
21        temp = heap[i];
```

OUTPUT

Enter size of first heap: Enter elements of first heap:

INPUT

4
90 70 50 40
3
85 60 30

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QUESTIONS 10 / 10 completed

Priority Queue

PRIORITY QUEUE

"Write a C program to perform the following operations using priority queue: Insert patient records with severity levels: Ravi-4, Kumar-2, Priya-5, Anu-1, Deepak-3 Display the patients in priority order Treat and remove the highest priority patient"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Patient
5 {
6     char name[20];
7     int priority;
8 };
9
10 struct Patient p[100];
11 int n = 0;
12
13 void swap(int a, int b)
14 {
15     struct Patient temp = p[a];
16     p[a] = p[b];
17     p[b] = temp;
18 }
19
20 void insert(char name[], int priority)
21 {
```

OUTPUT

INPUT

Your INPUT : here!

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QUESTIONS 10 / 10 completed

Heap- Min & Max

HEAP- MIN & MAX ELEMENT

Write a C program to find the maximum and minimum element using Heap.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i;
6     int arr[100];
7     int max, min;
8
9     printf("Enter number of elements: ");
10    scanf("%d", &n);
11
12    printf("Enter elements:\n");
13    for(i = 0; i < n; i++)
14    {
15        scanf("%d", &arr[i]);
16    }
17
18    max = arr[0];
19    min = arr[0];
20
21    for(i = 1; i < n; i++)
```

OUTPUT

INPUT

7
55 20 35 10

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QUESTIONS 10 / 10 completed

Min Heap- Operati

MIN HEAP- OPERATIONS

"Write a C program to construct a Min Heap using the following data: 55, 20, 35, 10, 90, 40, 15, 25 and perform the following operations:
Insert elements one by one into the Min Heap
Display the heap after every insertion
Delete the minimum element"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int heap[100];
4 int size = 0;
5
6 void insert(int value)
7 {
8     int i = size;
9     heap[size] = value;
10    size++;
11
12    while(i > 0)
13    {
14        int parent = (i - 1) / 2;
15
16        if(heap[parent] > heap[i])
17        {
18            int temp = heap[parent];
19            heap[parent] = heap[i];
20            heap[i] = temp;
21            i = parent;
```

OUTPUT

Min Heap after creation:

INPUT

Your INPUT here!